

Jeffery Patterson

Software Engineer · Los Angeles, California

designframebuildweb@gmail.com · 951-581-6263 · github.com/jpatterson933 · linkedin.com/in/jefferywpatterson · https://portfolio-production-c6b6.up.railway.app

Full-stack engineer with 4+ years of professional experience building scalable web applications, RESTful APIs, data visualization tools, and cross-platform desktop and mobile apps. Lifelong Lakers fan, been watching every game since Kobe's rookie year, with a deep curiosity about basketball analytics and the factors that drive organizational success in the NBA. Rapid learner who thrives on picking up new languages and frameworks to meet the needs of the team.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, C#, SQL, HTML, CSS	Frontend	React, Next.js, Vue (familiar), Tailwind CSS, TanStack, HTML5 Canvas
Backend	Node.js, Express, ASP.NET Core, REST APIs, BullMQ, Redis	Databases	PostgreSQL, MongoDB, Drizzle ORM, Prisma, Entity Framework Core
Cloud & DevOps	AWS, Docker, Railway, GitHub Actions, CI/CD, Sentry	Mobile / Desktop	Electron, Capacitor (iOS & Android), cross-platform deployment
Data & Analytics	pgvector, k6 / Grafana, Lighthouse CI, OpenAI, Anthropic, LangChain	Tooling	Git, Zod, Commander, Vitest, tsup, GraphQL, OAuth 2.0, MCP

EXPERIENCE

Full Stack Engineer Aug 2023 - Present
AE Studio · Los Angeles, CA

- Built and maintained internal web applications and client-facing platforms including data dashboards, assessment engines, webhook pipelines, and automation tooling
- Developed a cross-platform launcher (Electron desktop + Capacitor iOS/Android) with 50+ plugin integrations, demonstrating mobile and desktop deployment capability
- Architected a QTI 3.0-compliant assessment API with bidirectional JSON↔XML pipelines, load-tested to 5,000 virtual users via k6/Grafana, serving real students across the US in production
- Created data visualization and analytics tooling including bundle heat-maps, SEO/traffic dashboards, and AI-powered audit CLIs, turning raw data into actionable insights
- Implemented RESTful services backed by PostgreSQL and MongoDB with Redis caching, Drizzle ORM, and cloud deployment on AWS and Railway

Full Stack Development Teaching Assistant Feb 2023 - Aug 2023
UC San Diego & edX · Remote

- Mentored 30+ students through full-stack curriculum (React, Node.js, Express, MongoDB, MySQL, REST APIs) in an intensive bootcamp environment
- Guided students from concept to deployed web applications, reinforcing code review practices and Git workflows

SELECT PROJECTS

Universal Launcher
Cross-platform launcher deployed as Electron desktop app (Mac/Windows) and Capacitor mobile app (iOS/Android) with 50+ plugin integrations across Linear, Slack, GitHub, Notion, and Spotify.
React · TypeScript · Node.js · MongoDB · Electron · Capacitor

Timeback: QTI Assessment Engine
Production assessment API with bidirectional JSON↔XML pipelines across 17 interaction types, XSD validation, and multi-tenant MongoDB storage. Load-tested to 5,000 VUs.
TypeScript · Node.js · Express · MongoDB · k6 · Grafana

NBA Twitter Bot
Automated bot aggregating NBA news from 10+ sources via Perplexity AI, using OpenAI embeddings with cosine similarity for duplicate detection and GPT-4 for content generation.
Python · OpenAI · Perplexity AI · scikit-learn · Selenium

aiseo-audit (Open Source)
Deterministic CLI auditing web pages for AI search readiness across 7 categories and 30+ factors. Ships as both a CLI and typed API with JSON, Markdown, and HTML output.
TypeScript · Node.js · NLP · cheerio · Zod · Vitest

EDUCATION

UCLA, B.A. Political Science (2013 - 2016)
UCLA Extension, Full Stack Developer Bootcamp (2021)
UCLA Extension, Cybersecurity Certificate (2022 - 2023)
Santa Monica College, A.S. Computer Science (2022)